

Akhil Madineni

FULL-STACK SOFTWARE ENGINEER

Full-stack software engineer with 5+ years building enterprise web applications, React/Angular interfaces, Java/Spring services, and distributed workflow platforms. Currently contributing to Merino, a React/Java CI/CD platform orchestrated with Temporal across OCI, helping improve platform speed, reliability, and operations.



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WORK EXPERIENCE

Senior Member of Technical Staff | Oracle

JUNE 2025 - PRESENT

- Designed and implemented lock/unlock stage functionality across the React UI and Java backend, integrating stage controls with Temporal workflows.
- Built Terraform automation for Merino's control plane and data plane across all 51 OCI regions, provisioning the full regional stack and reducing new-region setup from days to about two hours per region.
- Implemented retry-safe workflow and queue-driven execution with backoff, hardening CI/CD runs and provisioning against transient failures.
- Built AI-assisted automation for pull-request reviews, automated log checks, and code-change generation for known security tickets, validated through tests and CI.
- Delivered zero-downtime, idempotent ADB password rotation for admin and application users using OCI Functions, Queues, Vault, and auditable logs/metrics/events, reducing manual operations and improving security.

Software Developer | Consult Veda

FEBRUARY 2025 - MAY 2025

- Built an observability system POC with Java, Spring Boot, and Kafka to ingest, process, and route telemetry through a reliable event-driven pipeline at scale.
- Developed J2EE/Spring ingestion and query REST APIs and integrated telemetry with dashboards and third-party systems, enabling faster debugging and clearer operational visibility.

Software Developer 3 | JPMorgan Chase

MAY 2023 - DECEMBER 2024

- Owned end-to-end delivery of a Splunk Universal Forwarder Generator, building its Angular UI, Spring Boot APIs, and Camunda-orchestrated provisioning workflow; reduced setup time by 30%.
- Built an EAC access workflow in Go for ACM, Datadog, Doppler, and Application EAC, reducing request processing time by 50%.

Software Developer | Walmart

OCTOBER 2022 - APRIL 2023

- Built Looper-based CI/CD automation and backend APIs for an internal observability dashboard with health/telemetry drilldowns, cutting release/deployment turnaround by about 30% and dashboard/query latency by about 20% through pipeline, API, and data-access optimizations.

Software Developer | Globant - Citi Bank

APRIL 2021 - SEPTEMBER 2022

- Built scalable QA automation for Citi/Globant by automating API test suites with BDD frameworks and Appium and designing a Brokerage Account Opening framework, reducing regression time by 30% and end-to-end onboarding time by 50%.



PROJECTS

Automated Essay Generation using Hugging Face Transformers | Kaggle Project

DECEMBER 2024

- Built a long-form generation pipeline with AutoTokenizer and AutoModelForCausalLM; improved quality and throughput with prompt templates, post-processing, batching, GPU acceleration, and automated pandas dataset/CSV submission preparation.

Brain Tumor Segmentation | Master's Project

AUGUST 2019 - DECEMBER 2020

- Built a brain MRI tumor segmentation model on BraTS 2019; implemented and compared U-Net and Octave U-Net with preprocessing and hyperparameter tuning, achieving Dice = 0.975.



TECHNICAL PUBLICATIONS | DZONE (2026)

- [Building Reliable Async Processing Pipelines Using Temporal](#) - Durable retries, compensation, recovery, and async orchestration.
- [A Low-Latency Routing Pattern for Multiple Small Language Models](#) - Cache-aware routing for low-latency multi-SLM serving.
- [A Practical Guide to Temporal Workflow Design Patterns](#) - Sagas, polling, parallelism, signals, and workflow versioning.
- [Orchestrating Zero-Downtime Deployments With Temporal](#) - Canary deployments with approvals, retries, and rollback.
- [Combine Temporal and Kafka for Resilient Distributed Systems](#) - Kafka event streaming with durable workflow state and recovery.
- [Multi-Scale Feature Learning in CNN and U-Net Architectures](#) - Multi-resolution feature fusion for CNN and U-Net segmentation.



EDUCATION

Master's in Computer Science | California State University, Sacramento

DECEMBER 2020

Bachelor's in Computer Science | GITAM University, Vizag, India

JUNE 2017



RELEVANT SKILLS

Front End: React, Angular, JavaScript, TypeScript, HTML5, CSS3

Backend & APIs: Java, J2EE, Spring Boot, REST APIs, Go, Kafka

Workflow, Cloud & DevOps: Temporal, Camunda, OCI, Terraform, Build Service, Pipelines, Functions, Queues, Vault, ADB, Looper, CI/CD

Observability, Testing & AI/ML: AI-assisted development, LLM coding tools, prompt engineering, Splunk, Datadog, Doppler, ACM, BDD, Appium, automated testing, Hugging Face Transformers, pandas, U-Net, Octave U-Net